

Modular Linear Equations Exercises

Q&A Notes

Question 1

Find all solutions to the equation

$$35x \equiv 10 \pmod{50}.$$

Answer 1

The solutions are

$$x \equiv 6, 16, 26, 36, 46 \pmod{50}.$$

Let

$$a = 35, \quad b = 10, \quad n = 50.$$

First compute

$$d = \gcd(35, 50) = 5.$$

Since

$$5 \mid 10,$$

solutions exist. There will be exactly

$$\boxed{5}$$

solutions modulo 50.

Now use Extended Euclid:

$$50 = 35(1) + 15,$$

$$35 = 15(2) + 5.$$

Back-substitute:

$$5 = 35 - 15(2).$$

Since

$$15 = 50 - 35,$$

we get

$$5 = 35 - 2(50 - 35) = 3(35) - 2(50).$$

Thus

$$5 = 35(3) + 50(-2),$$

so

$$x' = 3.$$

One solution is

$$x_0 = x' \left(\frac{b}{d} \right) \pmod{n}.$$

Therefore

$$x_0 = 3 \left(\frac{10}{5} \right) \pmod{50} = 6.$$

The spacing between solutions is

$$\frac{n}{d} = \frac{50}{5} = 10.$$

So the 5 solutions are

$$x_i = 6 + 10i \pmod{50}, \quad i = 0, 1, 2, 3, 4.$$

Hence

$$x \equiv 6, 16, 26, 36, 46 \pmod{50}.$$

Question 2

Prove that

$$ax \equiv ay \pmod{n}$$

implies

$$x \equiv y \pmod{n}$$

whenever $\gcd(a, n) = 1$. Show that the condition $\gcd(a, n) = 1$ is necessary by supplying a counterexample with $\gcd(a, n) > 1$.

Answer 2

If $\gcd(a, n) = 1$, then cancellation is valid modulo n .

From

$$ax \equiv ay \pmod{n},$$

we get

$$n \mid (ax - ay).$$

Factor:

$$n \mid a(x - y).$$

Since

$$\gcd(a, n) = 1,$$

Euclid's lemma gives

$$n \mid (x - y).$$

Therefore

$$\boxed{x \equiv y \pmod{n}}.$$

Now show why the condition is necessary. Take

$$a = 2, \quad n = 6, \quad x = 1, \quad y = 4.$$

Then

$$\gcd(2, 6) = 2 > 1.$$

Also,

$$ax = 2(1) = 2,$$

and

$$ay = 2(4) = 8 \equiv 2 \pmod{6}.$$

So

$$ax \equiv ay \pmod{6}.$$

But

$$1 \not\equiv 4 \pmod{6}.$$

Thus cancellation fails when $\gcd(a, n) > 1$.

Takeaway

$\boxed{\text{You may cancel } a \text{ modulo } n \text{ only when } \gcd(a, n) = 1.}$

Question 3

In the procedure `ModularLinearEquationSolver`, consider changing line 3 to

$$x_0 = x' \left(\frac{b}{d} \right) \bmod \left(\frac{n}{d} \right).$$

Will this work? Explain why or why not.

Answer 3

Yes, this change works. It prints the same solution set, possibly in a different order.

The original line computes

$$x_0 = x' \left(\frac{b}{d} \right) \bmod n.$$

The proposed line computes the same value modulo n/d :

$$x_0 = x' \left(\frac{b}{d} \right) \bmod \left(\frac{n}{d} \right).$$

This is valid because all solutions are spaced by

$$\frac{n}{d}.$$

If one solution is changed by a multiple of n/d , it is still one of the same solution family.

More directly, suppose x_0 is a solution. Let

$$x'_0 = x_0 - q \left(\frac{n}{d} \right)$$

for some integer q . Then

$$ax'_0 = ax_0 - q \frac{an}{d}.$$

Since $d \mid a$, the number a/d is an integer. Hence

$$q \frac{an}{d} = q \left(\frac{a}{d} \right) n$$

is a multiple of n . Therefore

$$ax'_0 \equiv ax_0 \pmod{n}.$$

So x'_0 is also a solution.

Example

For

$$14x \equiv 30 \pmod{100},$$

we have

$$d = 2, \quad x' = -7.$$

The old line gives

$$x_0 = (-7)(15) \bmod 100 = 95.$$

The changed line gives

$$x_0 = (-7)(15) \bmod 50 = 45.$$

The solution spacing is

$$\frac{100}{2} = 50.$$

So the old line prints

$$95, 45,$$

while the changed line prints

$$45, 95.$$

The same solutions appear.

Thus

the modified line works.

Question 4

Let p be prime and

$$f(x) \equiv f_0 + f_1x + \cdots + f_tx^t \pmod{p}$$

be a polynomial of degree t , with coefficients drawn from $\mathbb{Z}_p$. A value $a \in \mathbb{Z}_p$ is a zero of f if

$$f(a) \equiv 0 \pmod{p}.$$

Prove that if a is a zero of f , then

$$f(x) \equiv (x - a)g(x) \pmod{p}$$

for some polynomial $g(x)$ of degree $t - 1$. Prove by induction on t that a nonzero polynomial of degree t has at most t distinct zeros modulo p .

Answer 4

Because p is prime, $\mathbb{Z}_p$ behaves like a field: every nonzero element has a multiplicative inverse.

Part 1: Factor theorem modulo p

Divide $f(x)$ by $x - a$ in $\mathbb{Z}_p$. Polynomial division gives

$$f(x) \equiv (x - a)g(x) + r \pmod{p},$$

where r is a constant remainder.

Now substitute $x = a$:

$$f(a) \equiv (a - a)g(a) + r \pmod{p}.$$

So

$$f(a) \equiv r \pmod{p}.$$

Since a is a zero,

$$f(a) \equiv 0 \pmod{p}.$$

Therefore

$$r \equiv 0 \pmod{p}.$$

Hence

$$f(x) \equiv (x - a)g(x) \pmod{p}.$$

Since f has degree t , the polynomial g has degree $t - 1$.

Part 2: Root bound by induction

We prove: a nonzero polynomial of degree t has at most t distinct zeros modulo p .

Base case: If $t = 0$, then $f(x)$ is a nonzero constant. A nonzero constant is never 0 modulo p , so it has 0 roots. This is at most 0.

Inductive step: Assume every nonzero polynomial of degree $t - 1$ has at most $t - 1$ distinct zeros modulo p .

Let $f(x)$ have degree t . If f has no zeros, then the result is done. Otherwise, let a be one zero of f . By Part 1,

$$f(x) \equiv (x - a)g(x) \pmod{p},$$

where $g(x)$ has degree $t - 1$.

Now let $b \neq a$ be another zero of f . Then

$$0 \equiv f(b) \equiv (b - a)g(b) \pmod{p}.$$

Because $b \neq a$,

$$b - a \not\equiv 0 \pmod{p}.$$

Since p is prime, $b - a$ has an inverse modulo p . Therefore we can cancel $b - a$, giving

$$g(b) \equiv 0 \pmod{p}.$$

So every zero of f except a is a zero of g .

By the induction hypothesis, g has at most $t - 1$ zeros. Including a , the polynomial f has at most

$$1 + (t - 1) = t$$

zeros.

Therefore,

$$\boxed{\text{a nonzero polynomial of degree } t \text{ has at most } t \text{ distinct zeros modulo } p.}$$